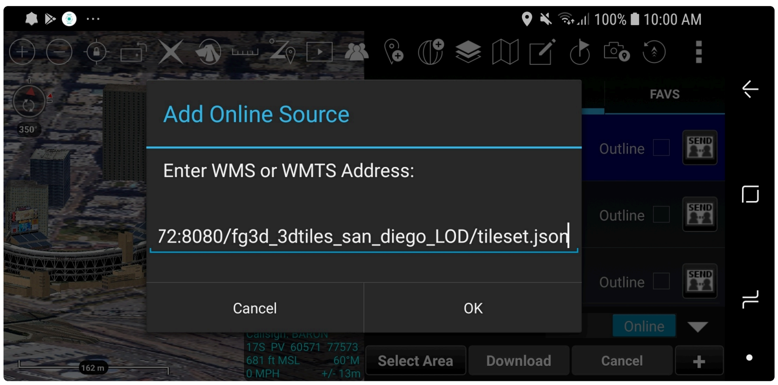
Cesium 3D Tiles

Created by Lawrence, Chris, last updated on Jul 04, 2020 • 3 minute read

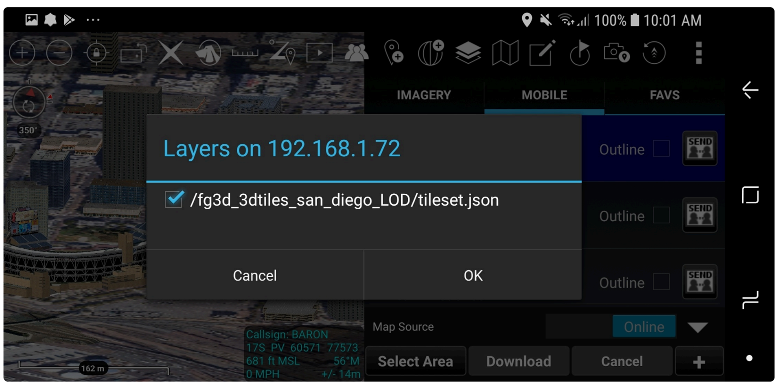
Connecting to a 3D Tiles Streaming Dataset

1. Launch the ATAK Map Manager
2. Go to the MOBILE tab.
3. Select the Down Arrow, then the Add (+) Button.

4. Type in the URL to the dataset's tileset.json file (e.g. `http://192.168.1.1:8080/samples/test_dataset_1/tileset.json`)



ATAK will send out a request to the specified URL to interrogate for available services (e.g. WMS, WMTS, WFS) and also check to see if it is is Cesium 3D Tiles tileset content. If the connection can be made, a window will display with the available services that were discovered.



5. Click the checkbox next to the tileset.

   ATAK will now import the tileset. A listing will be available under the Overlay Manager.

Local Cache

ATAK caches downloaded data under the directory

`atak/3dtilescache`

Each import has its own directory under that directory. Deleting the subdirectories or the `resources` directory is safe to do when ATAK is not running; ATAK will simply rebuild the cache next time it launches. Caches may also be transferred to other devices.

Setting up a Local HTTP Server for Testing Streaming

Cesium 3D Tiles datasets can be streamed over any HTTP/HTTPS connection.

Example Servers

Node.JS http-server (Recommended)

Mongoose Web Server

- https://github.com/cesanta/mongoose
- Direct Binary Download links https://www.cesanta.com/binary.html)

Using Node.JS http-server

Install NodeJS

- Download installer from https://nodejs.org/en/

Install `http-server`

From the command line, run

```
npm install --global http-server
```

Run `http-server`

The http-server application will serve content in the directory it is launched from over HTTP or HTTPS on the available network interfaces. Launch it on the command line by simply executing the command

```
http-server
```

Please review the documentation (https://www.npmjs.com/package/http-server) on *Usage* for enabling SSL connections and/or Basic Authentication.

An Android compatible CA certificate and private key for a http-server server should be generated with the following (note the subjectAltName with IP address is required to pass SSL peer verification on Android):

```
openssl req -newkey rsa:2048 -new -nodes -x509 -days 3650 -addext "subjectAltName = IP.1:XXX.XXX.XXX.XXX" -keyout key.pem -out cert.pem
```

Install/Removing Self Signed CA Certificates on Android

To load the CA certificate onto an Android device copy or download the certificate onto the device, then open the Settings app. Navigate to "Security → Advanced → Encryption & credentials → Install from storage" and select the .pem or .crt file.

You can remove the cert in the future by going to Settings app. Navigate to "Security → Advanced → Encryption & credentials → Trusted credentials" and select the "User" tab. Long press any item in the list to prompt removal.

Install Self Signed CA Certificates directly into ATAK

Generate a pkcs12 file for the certificate:

```
openssl pkcs12 -export -in cert.pem -inkey key.pem -out cert.p12
```

Copy or download the certificate onto the device. In ATAK open "Settings -> Network Connections -> Default SSL/TLS TrustStore Location" and select .p12 file.

WARNING: This method might interfere with other expected truststore. A better approach is to combine all needed certs into one p12 trustore.

No labels